Securing AI and Advanced Topics

Today's goal • 3/3 items

Module 1
Course Introduction

Module 2
Fraud Prevention with Cloud AI Solutions

Module 3
Introduction to Generative Adversarial Attacks (GANs)

Fundamentals of Generative Adversarial Networks (GANs)

- Introduction to Generative Adversarial Networks (GANs) Video • 8 min
- Reading References Reading • 15 min
- Fundamentals of Generative Adversarial Networks (GANs) Practice Assignment • Grade: 100%

Hands-On GAN Implementation and Synthetic Data Generation

- Getting to Know GANs Video • 8 min
- Reading References Reading • 10 min
- Hands-On GAN Implementation and Synthetic Data Generation Practice Assignment • Grade: 100%

Module-end Assessments

- **Self-Reflective Reading: Research GANs** Reading • 45 min
- Introduction to Generative Adversarial Attacks (GANs) Graded Assignment • 1h

Module 4
GANs and Adversarial Attacks

Understanding Black-box and White-box Adversarial Attacks

- Adversarial Attacks Explained Video • 8 min
- Reading References Reading • 10 min
- Understanding Black-box and White-box Adversarial Attacks Practice Assignment • 15 min

Practical Implementation of Adversarial Attacks

- Hands-On Adversarial Attacks Video • 9 min
- Reading References Reading • 10 min
- Practical Implementation of Adversarial Attacks Practice Assignment • 15 min

Module-end Assessments

- Self-Reflective Reading: GANs and Adversarial Attacks Reading • 45 min
- Practice Lab: Generating Synthetic QR Codes with the Trained Generator Lab • 1h
- GANs and Adversarial Attacks Graded Assignment • 1h

Module 5
Reinforcement Learning

Reinforcement Learning and Its Applications

- Reinforcement Learning Video • 8 min
- Reading References Reading • 10 min
- Reinforcement Learning and its Applications Practice Assignment • 15 min

Using RL for Adversarial Attacks and Optimizing Datasets

- Evading Malware Detectors with RL Video • 6 min
- Reading References Reading • 10 min
- Using RL for Adversarial Attacks and Optimizing Datasets Practice Assignment • 15 min

Module-end Assessments

- Self-Reflective Reading: Understanding Reinforcement Learning Reading • 45 min
- Reinforcement Learning Graded Assignment • 1h

Module 6
Evaluating AI Models and Performance

Feature Engineering and Model Optimization Techniques

- Challenges with Using AI for Cybersecurity Video • 8 min
- Reading References Reading • 10 min

Self-Reflective Reading: Research GANs

0:04 / 3:14 2x

Introduction

In this module, we explore the fascinating world of Generative Adversarial Networks (GANs), a powerful class of machine learning models that can generate realistic data by learning from existing datasets. We focus specifically on their application in creating adversarial examples, which can pose significant challenges in the field of cybersecurity.

The hands-on exercise involves several key tasks aimed at deepening your understanding of GANs, particularly in the context of generating adversarial examples. Below are the tasks you will need to complete, along with suggested steps to guide your exploration:

Tasks

1. **Literature Review on GANs**: Conduct a quick literature review on GANs. Describe the differences between the class example of emulating MNIST handwritten digits and the most current GANs used to create adversarial example images, network traffic, or malware.
2. **Impact of Accurate GANs on Cyber Analysts**: Discuss how the existence of accurate GANs for creating adversarial examples can negatively affect cyber analysts. Provide a real-world example to support your claims.
3. **Summary of GAN Development Process**: Use the language from the machine learning analytic development process to summarize the development of the GAN generator and discriminator in the MNIST handwritten digits class example.
4. **Performance Tuning of GAN Code**: Investigate the GAN code (generative_adversarial_network.py) from Chapter 8 in the provided virtual machine. Identify the performance tuning parameters and attempt to optimize these parameters to produce better results. Report how many epochs were necessary to achieve these results and explain how you know your results improved compared to those presented in the lecture.

Suggested Steps

1. Research GANs:

- Start by gathering relevant literature on GANs.
- Focus on recent advancements in the field to identify how current models differ from simpler examples like the MNIST generator.

2. Analyze Cybersecurity Implications:

- Reflect on the potential threats that advanced GANs pose to cybersecurity.
- Look for real-world incidents where GANs or similar technology have been utilized maliciously.

3. Review GAN Development:

- Examine the structure and functions of the GAN generator and discriminator.
- Use terminology related to machine learning analytics to articulate your understanding.

4. Optimize GAN Code:

- Open the provided GAN code and identify key parameters.
- Experiment with different settings and track the performance to see how your changes affect the results.

Self-Reflective Questions

- What new insights did you gain about the capabilities and applications of GANs in your literature review?
- How do you perceive the risks associated with the advancement of GAN technology in the context of cybersecurity?
- In what ways did optimizing the GAN parameters affect the quality of the generated images? What specific changes led to noticeable improvements?
- How might the knowledge gained from this module influence your approach to machine learning and cybersecurity in your future work?
- What challenges did you face while working with the GAN code, and how did you overcome them?

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